

Business Analysis Case Study: Bikes & Accessories Shop.

Business Context: The company aims to improve customer retention and increase revenue.

Goals of the analysis:

- Improve customer retention
- Increase revenue
- Optimize product performance

1. Executive Summary.

- Goal of the analysis: To understand why customers do not make repeat purchases and identify the key drivers of low retention.

- Key Insights:

Orders increased sixfold from winter 2012 to winter 2013. Customer retention also improved in 2013 compared to 2012. However, despite this growth, a significant share of customers (over 50%) falls into the “Lost Customers” category, indicating poor long-term retention.

- Recommendations:

- Strengthen customer retention through loyalty programs and personalized communication
- Focus on high-value segments (Big Spenders) to increase customer lifetime value
- Implement cross-selling strategies to boost revenue per customer
- Optimize regional strategy by reallocating resources to higher-performing markets

2. Introduction.

The main problem is low customer retention: only one third of customers make repeat purchases. To investigate this issue and identify the reasons behind low retention, I analyzed key performance indicators (KPI) and sales dynamics, and conducted RFM and cohort analysis using data from the end of 2010 to the beginning of 2014.

3. Methodology.

The data was sourced from the AdventureWorksDW2025 database in Microsoft SQL Server. SQL Server Management Studio 2022 was used to query and process the data.

The primary dataset was the FactInternetSales table, supplemented with dimension tables containing customer, product, date, and geography information.

I calculated key business metrics such as total revenue and average order value (AOV), performed customer segmentation, analyzed sales dynamics, identified top-performing products and regions, and conducted cohort-analysis to evaluate customer retention over time.

4. Findings & Visuals.

4.1 KPI and Customer Segmentation.

This section highlights the key metrics and customer behavior patterns relevant to the retention problem.

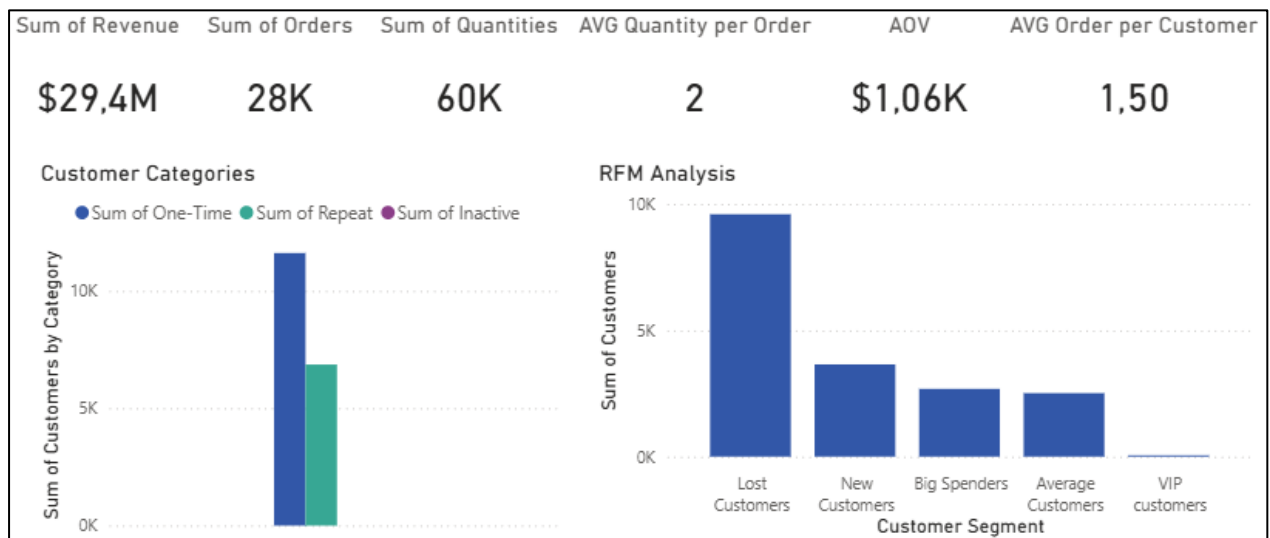
At the top of the dashboard, KPI metrics show that the average number of orders per customer is close to 1, indicating very low repeat purchase behavior. This suggests that most customers interact with the business only once.

Customer segmentation further confirms this pattern. Out of 18,484 customers, 62.8% are one-time buyers, while only 37.1% make repeat purchases. This means that repeat customers are significantly underrepresented, limiting long-term revenue growth.

RFM-analysis segments customers based on three key metrics: Recency (time since last purchase), Frequency (number of purchases) and Monetary Value (total spending). This approach helps identify customer groups such as VIP, lost, new, and other behavioral segments.

The share of “New Customers” is 19.8%, meaning that nearly every fifth customer has been recently acquired. However, there is a significant imbalance between “Lost Customers” (51.9%) and “VIP Customers” (0.0019%), indicating that most new customers do not convert into loyal ones and tend to churn after one or two purchases.

At the same time, “Average” and “Big Spenders” account for 28.1% of the customer base. These segments demonstrate clear purchase intent and engagement but have not yet transitioned into high-value loyal customers.



4.2 Cohort Analysis

For the cohort analysis, the period from 2012 to early 2014 was selected. Although the dataset includes earlier data starting from 2011, the business activity during that year was

very low. Starting from 2012, there is a significant increase in orders and revenue, providing a more stable and representative dataset for analyzing customer retention and behavior.

The cohort table shows minimal activity in 2012, with retention rates between 0.1% and 0.4%, indicating that these purchases were mostly sporadic and not part of consistent customer engagement. Loyalty in this period is nearly zero, suggesting that early customers did not establish recurring purchasing patterns.

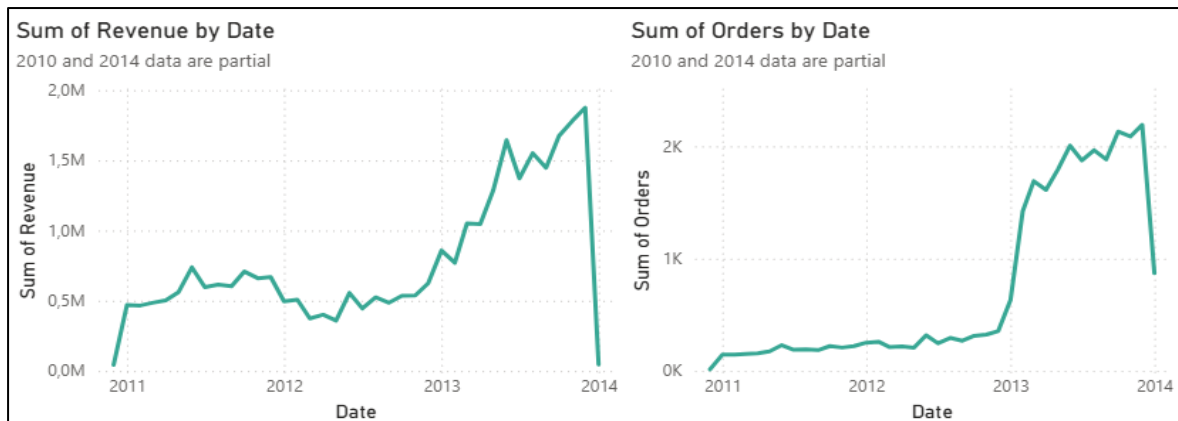
From February 2013 onwards, new cohorts show significant growth, with retention increasing from 1.5% to 6.7% in the first month. However, these customers rarely return after their initial purchase, and retention quickly drops to nearly 0.2%. This pattern indicates that while customer acquisition is effective, the products or experience do not sufficiently encourage repeat purchases.

Furthermore, the lifetime value (LTV) of 2013 cohorts is shorter than that of 2012 cohorts, as customer activity effectively disappears after six months. This highlights a key retention issue: although initial engagement is improving, sustaining long-term loyalty remains a challenge.

Cohort Analysis: retention %																								
Year	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
2012																								
January	1,4%											0,0%	0,0%	0,0%	0,1%	0,3%	0,1%	0,1%	0,1%	0,2%	0,2%	0,2%	0,2%	0,0%
February	1,4%										0,0%	0,0%	0,0%	0,0%	0,1%	0,3%	0,0%	0,0%	0,1%	0,2%	0,2%	0,2%	0,1%	
March	1,2%									0,0%	0,1%	0,0%	0,0%	0,1%	0,2%	0,1%	0,1%	0,1%	0,1%	0,3%	0,2%	0,0%		
April	1,2%									0,0%	0,0%	0,1%	0,1%	0,2%	0,1%	0,0%	0,1%	0,1%	0,2%	0,3%	0,2%			
May	1,1%							0,0%	0,0%	0,0%	0,1%	0,1%	0,1%	0,3%	0,0%	0,2%	0,1%	0,1%	0,1%	0,2%				
June	1,7%						0,0%	0,1%	0,0%	0,1%	0,0%	0,1%	0,2%	0,2%	0,1%	0,2%	0,1%	0,2%	0,5%					
July	1,4%						0,1%	0,0%	0,1%	0,0%	0,1%	0,2%	0,3%	0,2%	0,1%	0,1%	0,1%	0,3%						
August	1,6%				0,0%	0,1%	0,1%	0,1%	0,1%	0,1%	0,2%	0,2%	0,3%	0,2%	0,1%	0,1%	0,2%							
September	1,5%			0,0%	0,1%	0,1%	0,1%	0,1%	0,1%	0,2%	0,1%	0,3%	0,3%	0,2%	0,1%	0,1%								
October	1,7%		0,0%	0,1%	0,0%	0,1%	0,0%	0,1%	0,3%	0,1%	0,1%	0,4%	0,3%	0,1%	0,2%									
November	1,8%	0,0%	0,0%	0,0%	0,1%	0,0%	0,1%	0,2%	0,2%	0,1%	0,1%	0,4%	0,3%	0,1%	0,2%									
December	1,9%	0,0%	0,1%	0,1%	0,0%	0,1%	0,1%	0,2%	0,1%	0,1%	0,2%	0,4%	0,3%											
2013																								
January	2,7%	0,1%	0,1%	0,2%	0,2%	0,3%	0,1%	0,1%	0,1%	0,1%	0,2%	0,1%	0,1%											
February	6,9%	0,3%	0,3%	0,3%	0,3%	0,4%	0,3%	0,3%	0,4%	0,4%	0,3%	0,3%												
March	7,6%	0,2%	0,3%	0,2%	0,4%	0,3%	0,3%	0,3%	0,2%	0,2%	0,2%													
April	6,8%	0,2%	0,2%	0,2%	0,3%	0,2%	0,3%	0,2%	0,2%	0,2%														
May	6,8%	0,2%	0,2%	0,2%	0,2%	0,2%	0,2%	0,2%	0,2%															
June	7,4%	0,2%	0,2%	0,2%	0,3%	0,2%	0,2%	0,1%																
July	6,8%	0,2%	0,2%	0,2%	0,2%	0,3%	0,1%																	
August	6,8%	0,1%	0,2%	0,2%	0,2%	0,1%																		
September	6,4%	0,1%	0,2%	0,2%	0,1%																			
October	7,0%	0,1%	0,2%	0,2%																				
November	6,9%	0,2%	0,1%																					
December	7,2%	0,1%																						
2014																								
January	2,8%																							

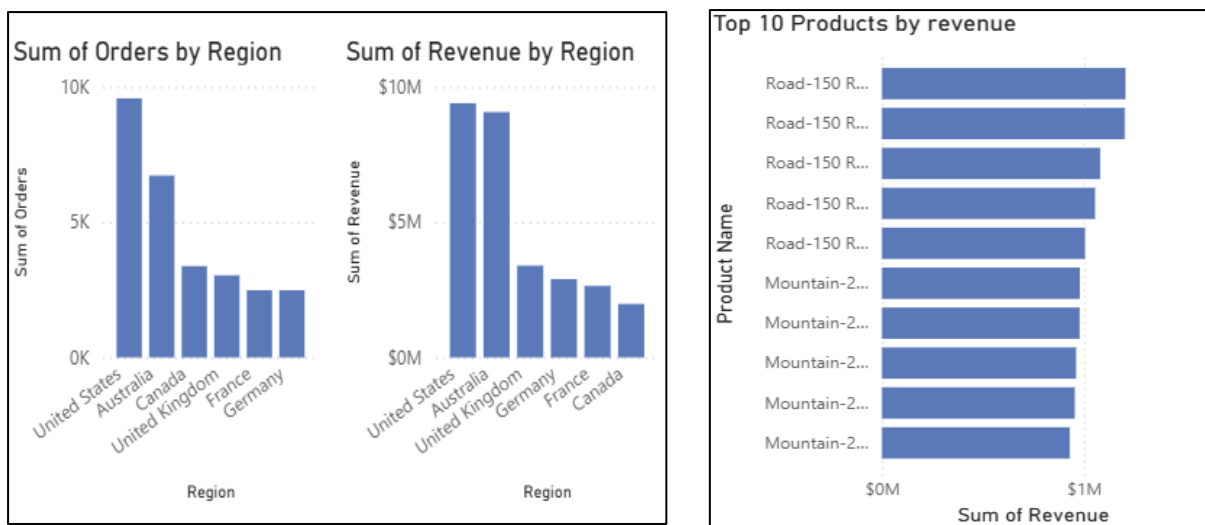
4.3 Sales Dynamics, Regional Distribution, and Top Products

Dynamic analysis of sales shows that both order numbers and revenue increased steadily from December 2012 to December 2013, peaking at the end of 2013. This indicates that there is no clear seasonality, as high sales activity is observed consistently throughout both summer and winter. Therefore, the high proportion of lost customers cannot be explained by seasonal fluctuations but rather reflects generally low customer loyalty.



Geographically, most of the revenue comes from the US and Australia. The US accounts for 32% of total revenue, while Australia contributes 30.9%. However, the number of orders in Australia is 1.4 times lower than in the US, suggesting that the higher revenue per order in Australia may be driven by delivery costs or logistics fees. Canada ranks third in order volume but last in revenue, representing 12.2% of orders and only 6.7% of revenue, which may indicate smaller average order value or price sensitivity in that market.

Analysis of product performance shows that the top 10 products are all bikes. Two specific bike models dominate sales, differing by variants and colors, and together they account for 3.53% of total revenue. This indicates a clear customer preference for these models, which could be leveraged in future marketing or pricing strategies.



5. Discussion and Insights

Core Issue: The business is growing extensively (through new customer acquisition) rather than intensively (through customer retention and loyalty).

- Sales Dynamic Insight.

Revenue increased threefold, and the number of orders grew sixfold from December 2012 to December 2013, while the average order value (AOV) declined. This indicates that growth is

driven by higher sales volume, but likely through lower-priced products or significant discounting, which may negatively impact profitability.

- RFM Insight.

“Big Spenders” accounts for 28.1% of the customer base, representing significant untapped potential. These customers generate high revenue but do not transition into VIP segments and show signs of declining engagement based on recency. This suggests missing opportunities in retention and customer development.

- Product Insight.

Top-performing products are exclusively bikes, which are typically purchased infrequently. If customers purchase a bike but do not return, it indicates a failure to cross-sell complementary products such as helmets, accessories, or maintenance services. This limits customer lifetime value.

- Geographic Insight.

Australia generates nearly the same revenue as the US despite having 1.4 times fewer orders. This suggests a higher revenue per order, potentially driven by higher-priced products, pricing strategy, or additional costs such as delivery and logistics.

Overall, these insights indicate that while demand generation is effective, the business lacks strategies to retain customers, increase purchase frequency, and maximize customer lifetime value.

6. Recommendations

1. Improve Customer Retention through Personalization and Engagement

To increase customer retention, the company should implement loyalty programs, personalized offers, and targeted communication strategies. This includes email campaigns, push notifications, and retargeting customers who show signs of inactivity. Personalized engagement can help re-activate customers and encourage repeat purchases.

2. Focus on High-Value Segments (Big Spenders)

The company should prioritize retention of “Big Spenders,” as retaining existing customers is significantly more cost-effective than acquiring new ones. Targeted strategies such as exclusive promotions, loyalty programs, or early access to sales can help convert these customers into loyal VIPs and increase their lifetime value.

3. Implement Cross-Selling Strategies

To increase customer lifetime value, the business should introduce cross-selling mechanisms. This can include offering accessories (e.g., helmets, gloves) shortly after the initial purchase, bundling bikes with complementary products, or providing discounts on first maintenance services. This approach can increase both revenue per customer and engagement.

4. Optimize Regional Strategy

Further analysis is needed to understand the low average order value in Canada. The company should evaluate pricing, product mix, and logistics in this region to improve profitability. At the same time, given the higher revenue per order in Australia, reallocating part of the marketing budget from lower-performing regions (such as Canada) to higher-performing ones may improve overall returns.